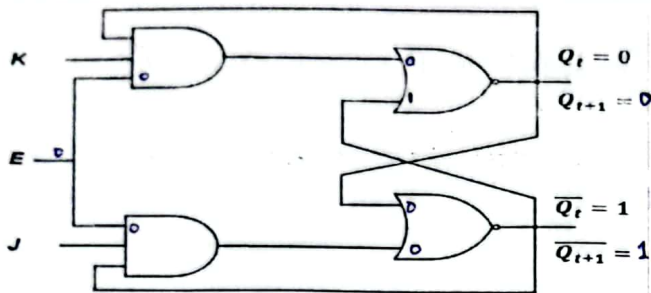
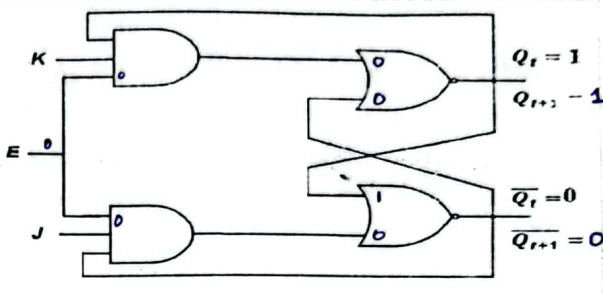
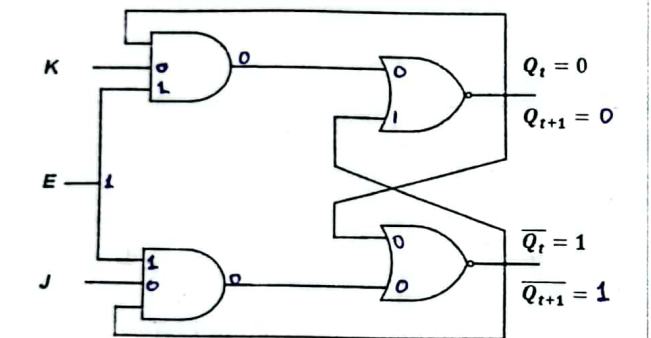
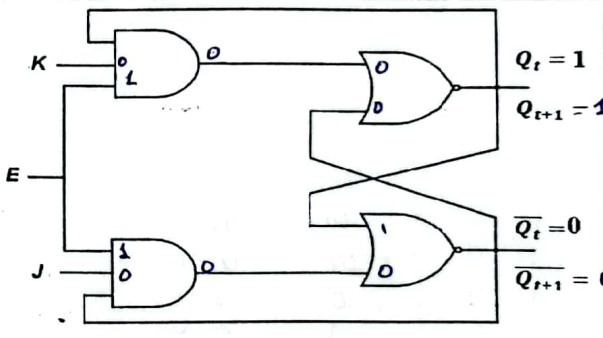
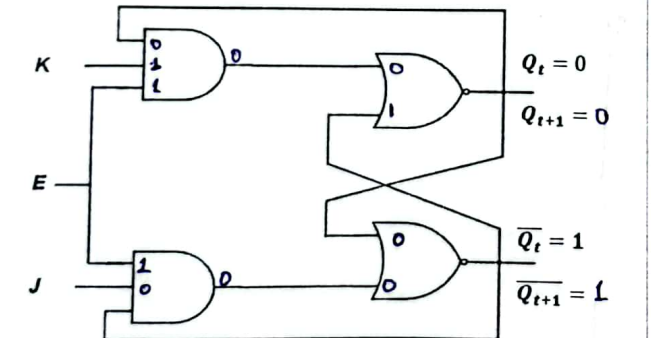
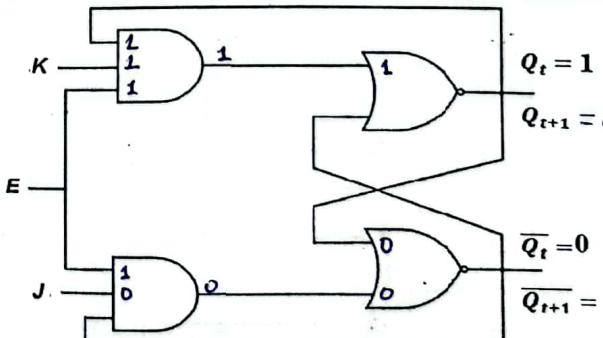


Instruction: No marks will be awarded for Direct Answers

1. Process the JK Latch given below for all possible cases and fill the table accordingly?

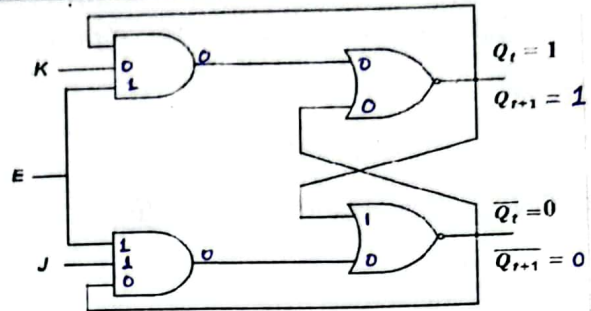
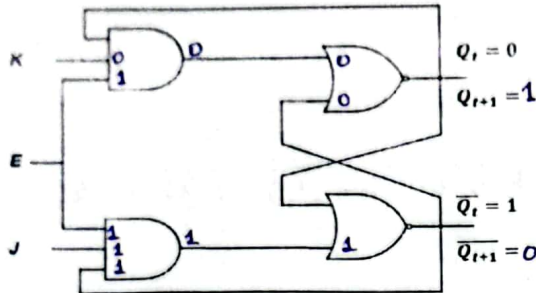
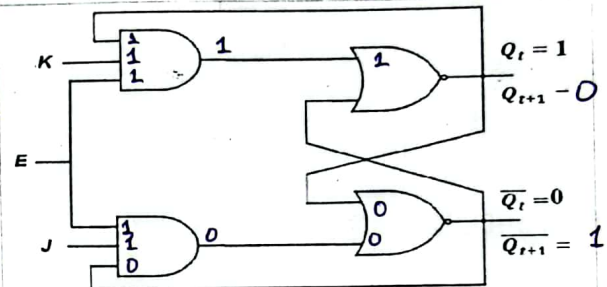
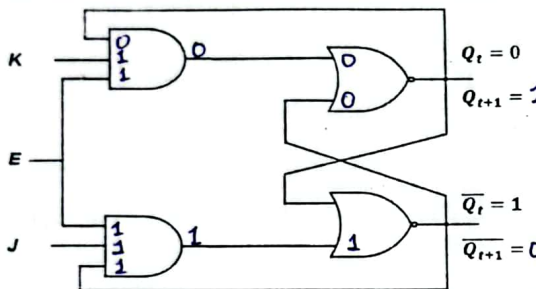
CIRCUIT:

Case-I ($Q_t = 0$)	Case-II ($Q_t = 1$)
$E=0, J=, K=, Q_t=0, Q_{t+1} = \underline{Q_t}$ 	$E=0, J=, K=, Q_t=1, Q_{t+1} = \underline{Q_t}$ 
$E=1, J=0, K=0, Q_t=0, Q_{t+1} = \underline{Q_t}$ 	$E=1, J=0, K=0, Q_t=1, Q_{t+1} = \underline{Q_t}$ 
$E=1, J=0, K=1, Q_t=0, Q_{t+1} = \underline{0}$ 	$E=1, J=0, K=1, Q_t=1, Q_{t+1} = \underline{0}$ 

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QUIZ 7-A

SECTION: F
 $E=1, J=1, K=0, Q_t=0, Q_{t+1}=1$
 $E=1, J=1, K=0, Q_t=1, Q_{t+1}=1$

 $E=1, J=1, K=1, Q_t=0, Q_{t+1}=\bar{Q}_t$
 $E=1, J=1, K=1, Q_t=1, Q_{t+1}=\bar{Q}_t$


JK Latch					JK LATCH with SET/RESET			
E	J	K	Q_{t+1}	$\bar{Q}_{t+1}$	SET	CLEAR	Q_{t+1}	$\bar{Q}_{t+1}$
0	X	X	Q_t	$\bar{Q}_t$	0	0	Q_t	$\bar{Q}_t$
1	0	0	Q_t	$\bar{Q}_t$	0	1	0	1
1	0	1	0	1	1	0	1	0
1	1	0	1	0	1	1	invalid	
1	1	1	$\bar{Q}_t$	Q_t				

2. Modify given NOR Based JK Flip Flop circuit to introduce Asynchronous CLEAR and SET inputs fill table above?

